

Plasmon-mediated pairing in 2D lives at small scattering angles; without them a critical coupling $\alpha_c \sim 1$ appears

Summary

The plasmon-pole interaction acts only for momentum transfers inside a window $q < K = a\kappa$ ($\kappa = N\alpha k_F$ the screening momentum, $a \lesssim 1$). At weak coupling every such transfer is a small scattering angle,

$$\phi \leq \phi_K \equiv \frac{K}{k_F} = aN\alpha \ll 1.$$

The argument has three steps.

1. The naive (BCS/Eliashberg ladder) gap equation has a solution at any α , $T_c \sim \Omega_K e^{-c^2/\alpha^2}$, and this solution is built entirely from angles below ϕ_K (Section 1).
2. Scattering at angles below $\phi_0 = (k_F R_s)^{-1/2}$ is the forward-scattering sector that multidimensional bosonization treats exactly. There the pair correlator acquires only a finite factor and no logarithm, so the ladder's Cooper logarithm from this sector is spurious, and the sector is removed from the gap equation (Section 2). *This is the one physical assumption of the note.*
3. With angles below ϕ_0 removed, the gap equation has no solution for $\alpha < \phi_0/Na$ (the plasmon window lies entirely below ϕ_0), and numerically requires $\alpha \gtrsim 1$ (Section 3). For $R_s \sim 1/\kappa$ this gives $\alpha_c \gtrsim 1/Na^2$.

Numerical checks and technical details are collected in the Appendix.

1. The naive gap equation and the angles it uses

Setup. $\alpha = e^2/\epsilon_\infty v_F$, N flavours, $\kappa = N\alpha k_F$, plasmon $\Omega_q^2 = \frac{1}{2}\kappa v_F^2 q$. The plasmon-pole interaction, valid for $|\nu| > v_F q$, is

$$U(i\nu, q) = \frac{2\pi e^2}{\epsilon_\infty q} \frac{\nu^2}{\nu^2 + \Omega_q^2} \theta(|\nu| - v_F q), \quad q < K = a\kappa, \quad (1)$$

and the linearized $T = 0$ gap equation ($\mathbf{k}$ on the Fermi surface) is

$$\Delta(w) = - \int \frac{dw'}{2\pi} \int \frac{d^2 p}{(2\pi)^2} U(w - w', |\mathbf{k} - \mathbf{p}|) \frac{\Delta(w')}{w'^2 + \xi_p^2}. \quad (2)$$

Units $v_F = \kappa = 1$: then $e^2/\epsilon_\infty = \alpha$, $\Omega_q^2 = q/2$, $K = a$, $\Omega_K^2 = a/2$, and $k_F = 1/N\alpha \gg 1$.

1.1 Reduction to a one-dimensional q -integral. Write $p = k_F + \xi$ and let ϕ be the angle between $\mathbf{p}$ and $\mathbf{k}$. For small angles $q^2 \simeq \xi^2 + k_F^2 \phi^2$ and $d^2 p \simeq k_F d\xi d\phi$. At fixed ξ substitute $\phi \rightarrow q$ ($k_F d\phi = q dq/\sqrt{q^2 - \xi^2}$, two branches $\pm\phi$), exchange the order of integration and set $\xi = q \sin \theta$:

$$\int \frac{d^2 p}{(2\pi)^2} \frac{U(\nu, q)}{w'^2 + \xi^2} = \frac{1}{2\pi^2} \int_0^K dq q U(\nu, q) \int_{-q}^q \frac{d\xi}{(w'^2 + \xi^2)\sqrt{q^2 - \xi^2}} = \frac{1}{2\pi} \int_0^K dq \frac{q U(\nu, q)}{|w'| \sqrt{w'^2 + q^2}}. \quad (3)$$

The condition $|\nu| > q$ involves only the full transfer q , so it just lowers the upper limit to $q_{\max}(\nu) = \min(K, |\nu|)$. With $qU = 2\pi\alpha \nu^2/(\nu^2 + \Omega_q^2)$:

$$\Delta(w) = - \frac{\alpha}{2\pi} \int dw' \int_0^{q_{\max}(\nu)} dq \frac{\nu^2}{\nu^2 + \Omega_q^2} \frac{\Delta(w')}{|w'| \sqrt{w'^2 + q^2}}, \quad \nu = w - w'. \quad (4)$$

1.2 What q means in terms of angles. At fixed q the angle is $|\phi| = \sqrt{q^2 - \xi^2}/k_F$: it equals q/k_F on shell ($\xi = 0$) and goes to zero for $|\xi| \rightarrow q$. Two consequences:

- (a) Every term of (4) has $|\phi| \leq q/k_F \leq \phi_K = aN\alpha$, on shell and off shell alike. The whole equation is small-angle scattering.
- (b) After the ξ -integration a given q contains *all* angles from 0 to q/k_F . A lower cut on q is therefore not a lower cut on the angle; an angular cut has to be imposed before the ξ -integration (Section 3.1).

1.3 The solution is built from the smallest angles. The kernel of (4) is positive, so a solution must change sign in frequency (Tolmachev–Anderson). Projecting on a low-frequency block (below $\simeq 0.3\Omega_K$) and a high-frequency block gives the condition $\alpha^2 M \ln(\Omega_K/E) \simeq 1$, with E the infrared scale: a solution exists for any α , with $T_c \sim \Omega_K e^{-c^2/\alpha^2}$ ($c \simeq 2.7$ at $a = 0.3$; previous note). The logarithm is the Cooper logarithm of transfers inside the window, i.e. of angles below ϕ_K . Figure 1b makes this quantitative: removing the angles $|\phi| < q_f/k_F$ (Section 3.1) and recomputing the pairing eigenvalue μ : removing the angles below $0.1\phi_K$ already reduces $|\mu|$ by 35%, below $0.45\phi_K$ by a factor 4.5, below $0.85\phi_K$ by a factor 30.

2. Why the small-angle sector is removed

The Cooper ladder is the correct resummation (equivalent to one-loop RG in the Cooper channel) for an interaction that transfers momenta of order k_F . Equation (4) is a ladder for an interaction whose transfers are all $\leq aN\alpha k_F$. For such forward scattering an exact treatment exists, and it gives a different answer.

2.1 The forward-scattering sector. Let R_s be the range of the interaction ($q \lesssim 1/R_s$). Divide the Fermi surface into patches and linearize the dispersion on each: a state at angle ϕ from the patch centre has the curvature energy $k_F^2 \phi^2 / 2m$, which is negligible against the energy scale v_F/R_s probed by the interaction when

$$\phi \ll \phi_0 \equiv \frac{1}{\sqrt{k_F R_s}}. \quad (5)$$

Inside such flat patches the density on each patch is a free boson and the interaction is quadratic in the bosons: the sector of scattering angles $|\phi| \lesssim \phi_0$ is exactly solvable (multidimensional bosonization). For $R_s \sim 1/\kappa$, $\phi_0 \sim \sqrt{N\alpha}$, parametrically larger than $\phi_K = aN\alpha$.

2.2 The pair correlator in this sector. A fermion on patch n is $\psi_n \propto e^{i\Phi_n}$, so the pair operator on opposite patches, $P = \psi_n \psi_{\bar{n}}$, has the exact correlator

$$\chi_P(x, \tau) = \chi_P^{(0)}(x, \tau) e^{-Q_P(x, \tau)}, \quad Q_P = Q_{nn} + Q_{\bar{n}\bar{n}} + 2Q_{n\bar{n}}, \quad (6)$$

$$Q_{nm}(x, \tau) = \int \frac{d^2 q}{(2\pi)^2} \frac{d\nu}{2\pi} \frac{V(q, i\nu) [1 - \cos(\mathbf{q} \cdot \mathbf{x} - \nu\tau)]}{(i\nu - \mathbf{v}_n \cdot \mathbf{q})(i\nu - \mathbf{v}_m \cdot \mathbf{q})}. \quad (7)$$

The question is whether Q_P grows logarithmically at large x, τ (which would be the counterpart of the Cooper logarithm) or saturates. In the plasmon window $|\nu| > q$ both denominators are $\simeq -\nu^2$, the four terms add, and at large distance

$$|Q_P^{(\text{pl})}| \simeq \frac{4\alpha}{\pi} \int_0^K \frac{dq}{\Omega_q} \arctan \frac{\Omega_q}{q} \leq 4\sqrt{2} \alpha \sqrt{a}, \quad (8)$$

numerically $|Q_P| = 2.4\alpha$ for $a = 0.3$. The integral converges at small q because $\int d\nu/(\nu^2 + \Omega_q^2) \propto q^{-1/2}$. There is no logarithm: forward scattering in the plasmon window multiplies the pair bubble by a constant $1 + O(\alpha\sqrt{a})$.

2.3 Consequence, and what is assumed. The ladder over the same sector produces $\ln(\Omega_K/E)$ and $T_c \sim e^{-c^2/\alpha^2}$; the exact treatment produces a finite factor. We therefore remove the angles $|\phi| < \phi_0$ from the gap equation and keep the ladder only for $|\phi| > \phi_0$. Two points are not settled here: (i) the static part of the sector, $|\nu| < v_F q$, is not included in (8); for a static repulsion the known result is a power-law *suppression* of χ_P (Miserev, Schoeller, Klinovaja, Loss, arXiv:2312.17208), which cannot help pairing; (ii) ϕ_0 is defined only up to a factor of order one. Section 3 therefore treats ϕ_0 as a free parameter.

3. Gap equation without the small-angle sector

3.1 The angular cut. By 1.2(b) the cut $|\phi| > \phi_0$ must be imposed in the (ξ, ϕ) variables. With $q^2 = \xi^2 + k_F^2 \phi^2$ it reads $q^2 > \xi^2 + q_f^2$, where

$$q_f \equiv k_F \phi_0 = \frac{\phi_0}{N\alpha} \quad (\text{units of } \kappa). \quad (9)$$

Exchanging the order of integration, this is $q > q_f$ and $|\xi| < b(q) \equiv \sqrt{q^2 - q_f^2}$. The ξ -integral of (3) over the reduced range is again elementary:

$$\int_{-b}^b \frac{d\xi}{(w'^2 + \xi^2)\sqrt{q^2 - \xi^2}} = \frac{2}{|w'|\sqrt{w'^2 + q^2}} \arctan\left(\frac{\sqrt{w'^2 + q^2}}{|w'|} \frac{b}{q_f}\right). \quad (10)$$

This is exact within the small-angle kinematics (Appendix A.1), and gives $\pi/|w'|\sqrt{w'^2 + q^2}$ for $q_f \rightarrow 0$. The factor $\frac{2}{\pi} \arctan(\cdot)$ is the fraction of weight that survives the cut. In the Cooper regime $|w'| \ll q$ it is $\simeq 1$: the relevant states are on shell, and the angle cut acts like a plain cut $q > q_f$. In the plasmon regime $|w'| \gg q$ it is $\frac{2}{\pi} \arccos(q_f/q) < 1$: off-shell states at small angle are also removed. The gap equation becomes

$$\Delta(w) = -\frac{\alpha}{2\pi} \int dw' \int_{q_f}^{q_{\max}(\nu)} dq \frac{\nu^2}{\nu^2 + \Omega_q^2} \frac{2}{\pi} \arctan\left(\frac{\sqrt{w'^2 + q^2}\sqrt{q^2 - q_f^2}}{|w'|q_f}\right) \frac{\Delta(w')}{|w'|\sqrt{w'^2 + q^2}}. \quad (11)$$

Since the cut is fixed in angle, q_f depends on α through (9).

3.2 Exact bound: the window must be open. Equation (11) has an empty q -range unless $q_f < K$, i.e.

$$\alpha > \alpha_{\min}(\phi_0) = \frac{\phi_0}{Na} \quad (12)$$

For smaller α every plasmon transfer is an angle below ϕ_0 : the whole window lies in the forward-scattering sector and nothing is left for the ladder.

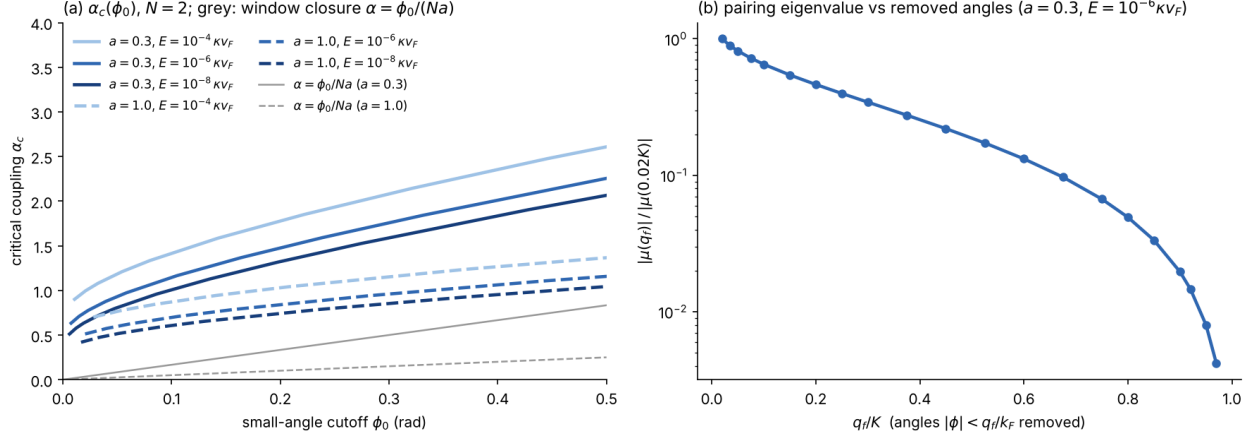


Figure 1: (a) Critical coupling of (11) versus the small-angle cutoff ϕ_0 ($N = 2$; two window sizes a , three infrared scales E), plotted parametrically, $\alpha_c = -1/\mu(q_f)$, $\phi_0 = N\alpha_c q_f$. Grey: the exact bound (12). (b) Pairing eigenvalue of the naive equation (4) with the angles $|\phi| < q_f/k_F$ removed, relative to $q_f = 0.02K$ ($a = 0.3$, $E = 10^{-6}\kappa v_F$).

3.3 Numerical solution. Equation (11) is written as $\Delta = -\alpha A \Delta$ and the lowest eigenvalue μ of A gives $\alpha = -1/\mu$. Since A depends on α only through q_f , we compute $\mu(q_f)$ and read the critical coupling from $\alpha_c = -1/\mu(q_f)$, $\phi_0 = N\alpha_c q_f$ (Fig. 1a and the table; $N = 2$, E in units of κv_F).

α_c ($N = 2$)	$\phi_0 = 0.05$	0.1	0.2	0.3
$a = 0.3, E = 10^{-4}$	1.19	1.42	1.78	2.08
$a = 0.3, E = 10^{-6}$	0.93	1.15	1.48	1.76
$a = 0.3, E = 10^{-8}$	0.81	1.01	1.32	1.59
$a = 1.0, E = 10^{-4}$	0.77	0.87	1.03	1.16
$a = 1.0, E = 10^{-6}$	0.60	0.70	0.84	0.96
$a = 1.0, E = 10^{-8}$	0.51	0.61	0.74	0.86
bound (12), $a = 0.3$	0.08	0.17	0.33	0.50
bound (12), $a = 1.0$	0.025	0.05	0.10	0.15

What the numbers say: (i) α_c is of order one for any reasonable ϕ_0 (0.6–1.4 at $\phi_0 = 0.1$). (ii) α_c grows slowly with ϕ_0 (roughly $\phi_0^{1/3}$) and decreases with the window size (roughly $a^{-1/2}$). (iii) α_c still decreases with E : for angles between ϕ_0 and ϕ_K the ladder is legitimate, the Cooper logarithm is real, and $\alpha_c \rightarrow \alpha_{\min}$ as $E \rightarrow 0$. But this happens only like $[\ln(\Omega_K/E)]^{-1/2}$: at $\phi_0 = 0.1$, $a = 0.3$ the bound is 0.17 while $\alpha_c = 1.0$ at $E = 10^{-8}$.

3.4 The screened Coulomb case. With $R_s \sim 1/\kappa$, $\phi_0 \sim \sqrt{N\alpha}$ and the bound (12) becomes a condition on α alone, $\alpha > \sqrt{N\alpha}/Na$, i.e.

$$\alpha_c \gtrsim \frac{1}{Na^2}, \quad (13)$$

which is 0.5 for $N = 2$, $a = 1$ and 5.6 for $a = 0.3$. Equivalently, at weak coupling $\phi_K = aN\alpha \ll \sqrt{N\alpha} = \phi_0$: the entire plasmon window is forward scattering.

Conclusion

The weak-coupling solution of the plasmon gap equation is a Cooper logarithm produced by forward scattering, $\phi \leq aN\alpha$. Treating that sector exactly instead of by the ladder removes it; what remains has no solution below $\alpha_{\min} = \phi_0/Na$ and in practice requires $\alpha_c \simeq 1$, where the one-loop treatment is no longer controlled. Plasmon-mediated superconductivity at weak coupling is excluded, given the assumption of Section 2.

Appendix: checks and numerical details

A.1 Angular cut, Eq. (10). Equation (10) inserted into (3) was compared with a direct two-dimensional integration over (ξ, ϕ) with $|\phi| > \phi_0$ imposed literally, with the small-angle kinematics and with the exact ones, $q^2 = \xi^2 + 2k_F(k_F + \xi)(1 - \cos \phi)$, measure $(k_F + \xi) d\xi d\phi$ ($k_F = 5\kappa$, `check_phi_cut.py`):

w'	ν	ϕ_0	$q_{\max}$	direct, exact	direct, small-angle	Eq. (10)	plain $q > q_f$
0.3	0.8	0.02	0.3	1.02463	1.02454	1.02454	1.605
0.2	0.9	0.05	0.3	0.24429	0.24368	0.24368	0.629
0.02	0.3	0.03	0.25	10.7335	10.7306	10.7306	12.24
0.01	0.5	0.02	0.3	76.5619	76.5479	76.5479	80.92
0.001	0.5	0.04	0.3	270.434	270.303	270.303	271.93

With the small-angle kinematics the agreement is to 10^{-15} ; the exact kinematics differ by at most 2.5×10^{-3} (curvature, relative order q/k_F). A plain lower cut on q is wrong by up to a factor 2.6 in the plasmon regime. In the gap equation the difference is quantitative: at $a = 0.3$, $E = 10^{-6}$, $q_f = 0.1K$ ($0.3K$) the angle cut gives $\alpha = 0.97$ (1.84), the q -cut 0.87 (1.47).

A.2 The plasmon condition. Replacing $\theta(|\nu| - v_F q)$ by the upper limit $q_{\max}(\nu)$ in (4) is exact, because the condition involves only the full transfer q and not ξ or ϕ separately. The condition is on the transfer frequency ν , not on w' : pairs with $|w'| \ll v_F q$ exchanging a plasmon with $|\nu| \gg v_F q$ are inside the validity domain.

A.3 Numerics. Logarithmic frequency grid from E to $10\Omega_K$ (34 points per decade, Δ even in w), Gauss-Legendre in q after $q = |w'| \sinh t$ (140 nodes); grid errors are below 10^{-4} . Raising the frequency cutoff from $10\Omega_K$ to $30\Omega_K$ changes μ by 3%. Table entries solve $\alpha \mu(\phi_0/Na) = -1$ directly at fixed ϕ_0 (`solve_angular_cutoff.py`, `make_alphac.py`).